

ClearEval: A Database-Grounded Feasibility Benchmark for LLM-Generated Tissue Optical Clearing Protocols

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Abstract—Tissue Optical Clearing (TOC) enables non-destructive, high-resolution 3D imaging, but designing a feasible protocol requires aligning the experimental goal, specimen type, and fluorescence/labeling modality with an appropriate clearing method and protocol parameters. Existing biomedical LLM evaluations have emphasized static knowledge, question answering, and text-based reasoning, while the experimental feasibility of generated wet-lab protocols remains less studied. We introduce ClearEval, a database-grounded, text-reference-free feasibility benchmark for LLM-generated TOC protocols. ClearEval is built on a structured TOC feasibility database integrating 372 wet-lab experimental records, 400+ curated tissue-clearing literature chunks, and expert-reviewed constraints on methods, tissues, markers, timing, and compatibility. It contains 253 open-ended protocol-generation scenarios for evaluating TOC protocols, plus a 2,588-question static knowledge audit to separate knowledge gaps from application failures. Generated protocols are evaluated with CCE, a three-part framework covering Completeness, Correctness, and Effectiveness. Completeness and Correctness are scored by LLM-based rubric judgment, while Effectiveness is computed from database-grounded feasibility rules and formulas. Applying ClearEval to 13 state-of-the-art LLMs, we find that static TOC knowledge does not reliably translate into feasible protocol generation. The dominant failures arise from executable parameterization in TOC, especially matching fluorescent markers to required labeling targets. ClearEval converts expert criteria for TOC protocol feasibility into a computable evaluation framework. It provides a depth-first feasibility benchmark for TOC and a re-curatable template for building similar benchmarks in other wet-lab domains. Code and data: <https://anonymous.4open.science/r/ClearEval>.

Index Terms—Benchmark, Tissue optical clearing, Wet-lab protocols, Large language models

I. INTRODUCTION

Tissue Optical Clearing (TOC) renders biological tissues optically transparent by reducing light scattering and refractive-index mismatch, enabling non-destructive, high-resolution 3D imaging via light-sheet microscopy [1], [2]. However, designing a TOC protocol is not a single-answer task. Given a specimen, labeling target, imaging goal, and laboratory constraint, experimenters must choose a feasible combination of clearing method, labeling strategy, refractive-index condition, incubation schedule, sample geometry, and operational constraints. Multiple combinations may be acceptable, but one incompatibility—for example, a quenched fluorophore, an unsuitable method for the sample scale, or an unrealistic

incubation time—can invalidate the protocol and irreversibly damage the specimen.

Existing biomedical LLM benchmarks, including MedQA [3], MMLU [4], and recent biomedical extensions [5]–[7], have advanced the evaluation of biomedical knowledge, question answering, and text-based reasoning. For generated wet-lab protocols, the key question is not only textual plausibility, but alignment with real experimental constraints. Likewise, semantic metrics such as BLEU [8] and general-purpose LLM evaluators [9], [10] do not explicitly evaluate method–label compatibility, refractive-index suitability, or timing plausibility. This creates a gap between knowledge evaluation and pre-experimental feasibility checking.

We introduce **ClearEval**, a database-grounded, text-reference-free feasibility benchmark that converts expert judgment for TOC protocol feasibility into a structured, computable framework. To make these criteria operational, we build a feasibility database from 372 wet-lab experimental records, 400+ curated tissue-clearing literature chunks, and expert-reviewed constraints on clearing methods, labeling, refractive-index matching, timing, and label compatibility. The database supports both the construction of protocol-design scenarios and the scoring of generated protocols. ClearEval contains 253 open-ended protocol-generation scenarios for evaluating TOC protocols, plus a 2,588-question static knowledge audit to separate knowledge gaps from application failures. Generated protocols are evaluated with CCE (Completeness, Correctness, Effectiveness): Completeness and Correctness are scored by LLM-based rubrics over free-form procedural text, while Effectiveness is computed from database-grounded checks of method suitability, label compatibility, refractive-index matching, and timing plausibility.

Experiments on 13 state-of-the-art LLMs show a clear gap between static TOC knowledge and protocol-level feasibility. Knowledge Index scores are high and close across models, while Application Index scores are lower and vary more across models (Table IV). The CCE results show that many protocols are complete and logically organized, but fail on experimental feasibility checks. The main error is in executable parameter choices, especially matching fluorescent markers to the required labeling targets.

Our contributions are: (1) a structured TOC feasibility

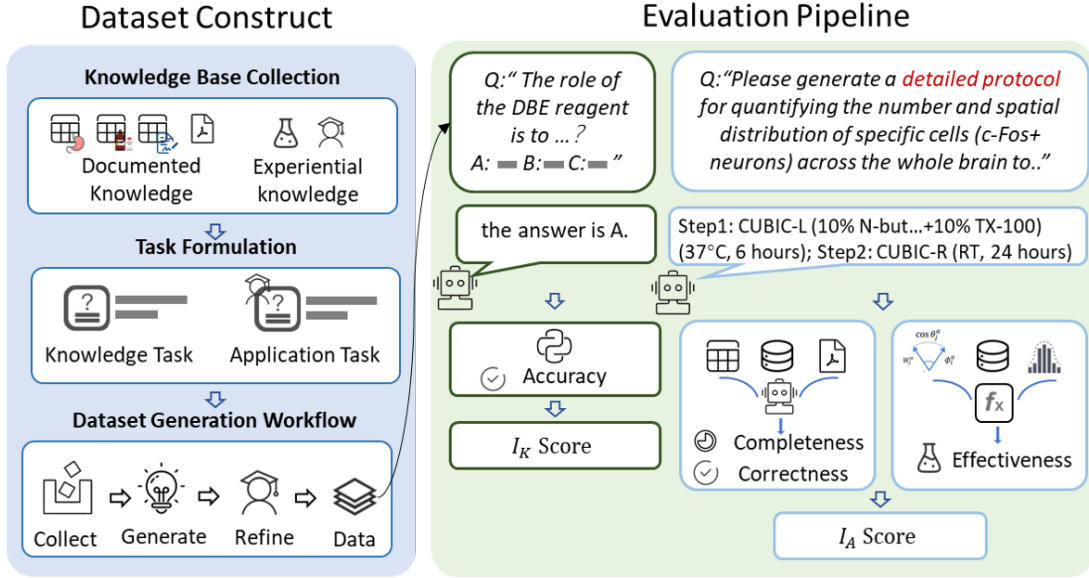


Fig. 1. Overview of ClearEval. A TOC feasibility database is constructed from wet-lab experimental records, curated literature chunks, and expert-reviewed constraints. The database supports both task construction and protocol scoring. The Knowledge track is scored by the Knowledge Index (I_K), while the Application track is scored by the Application Index (I_A), which combines Completeness, Correctness, and Effectiveness.

database built from experimental records, literature evidence, and expert-reviewed rules; (2) a text-reference-free benchmark for evaluating LLM-generated TOC protocols, with a static knowledge audit for diagnosing knowledge gaps; and (3) a database-grounded CCE evaluator that turns TOC expert judgment into computable checks for protocol feasibility.

II. RELATED WORK

Scientific and biomedical LLM evaluation. Biomedical LLM benchmarks have grown from general knowledge tests (MMLU [4], MedQA [3], Med-PaLM [11]) to medical and biological tasks [5]–[7] covering domain knowledge, clinical reasoning, and question answering. ClearEval instead evaluates whether the experimental choices in a free-text, LLM-generated TOC protocol are feasible.

Biological protocol and wet-lab benchmarks. Recent benchmarks cover biological protocols and lab workflows: Bio-ProBench [12] (protocol QA, step ordering, error correction, generation), LAB-Bench [13] (literature use, planning, data analysis), BixBench [14] (computational-biology agents), and SeedBench [15] (seed science). ClearEval targets Tissue Optical Clearing and scores feasibility at the parameter level (clearing method, marker–target matching, refractive index, timing); because multiple TOC routes can be valid, it uses database-grounded checks rather than similarity to a single reference.

Automatic evaluation, TOC, and pre-experimental checking. Automatic evaluators range from text-overlap metrics (BLEU [8]) to LLM judges (G-Eval [9]), whose bias and consistency are actively studied [10], [16]. ClearEval uses LLM judgment only for Completeness and Correctness, and scores Effectiveness with non-LLM rules grounded in the feasibility database. TOC methods such as CLARITY, CUBIC,

iDISCO, uDISCO, and PEGASOS support high-resolution 3D imaging across scales [1], [2], [17]–[19]; ClearEval encodes the relations among method, scale, labeling, refractive index, and time from this literature into a structured database for pre-experimental checking.

III. THE CLEAREVAL BENCHMARK

ClearEval has three parts: a TOC feasibility database, a task suite, and a scoring framework (Fig. 1). The database stores experimental records, literature evidence, and expert-reviewed TOC rules in a machine-readable form. It is used to build protocol-design scenarios and to score generated protocols. The task suite contains open-ended TOC protocol scenarios, together with a static knowledge audit used to separate knowledge gaps from application failures. The scoring framework reports a Knowledge Index (I_K) for the knowledge audit and an Application Index (I_A) for generated protocols.

A. TOC Feasibility Database

The TOC feasibility database is built to make expert judgment on protocol feasibility checkable. It is constructed from 372 wet-lab experimental records, 400+ curated tissue-clearing literature chunks, and expert-reviewed rules on tissues, molecular targets, fluorescent markers, clearing methods, refractive-index conditions, sample scales, and compatibility. These materials are organized into seven machine-readable knowledge bases. Table I summarizes their content, sources, and use in the Effectiveness score; full curation details are provided in Appendix A.

B. Task Suite Construction

ClearEval contains a Knowledge audit and an Application track. The Knowledge audit is used to check whether a model

TABLE I
FEASIBILITY DATABASE INVENTORY. EACH KNOWLEDGE BASE STORES
TOC EVIDENCE OR EXPERT-REVIEWED RULES USED FOR TASK
CONSTRUCTION OR PROTOCOL SCORING.

Knowledge Base	Entries	Source
Method capability vectors	20 methods, 6 axes	Expert + lit.
Fluorophore compat. matrix	marker $\times$ method	Expert + lit.
Tissue & target table	30+ tissue types	Lit. + expert
RI references (σ_{RI})	20 clearing methods	Literature
Timing database (τ)	126 (method, tier) rows	Wet-lab + lit.
Wet-lab experimental records	372 protocols	Lab data
Literature chunks	400+ paragraphs	Literature

has basic TOC knowledge before its protocol-generation failures are interpreted. We start from expert-written templates and instantiate them with terms from the feasibility database. This process produces 2,588 multiple-choice questions, including 533 named-entity questions, 1,350 domain-knowledge questions, and 705 task-scenario questions. Each question has eight options, followed by deduplication and option shuffling. Reported I_K values are computed on a 768-question development split, with 30 questions held out for validation.

The Application track evaluates generated TOC protocols. We define 60 expert-designed scientific missions across biological scales and combine them with the feasibility database to build constrained experimental scenarios. We then add real wet-lab challenges and filter the set to 253 open-ended tasks. These tasks are evaluated by feasibility rather than reference-text similarity, because more than one protocol route can be valid. The scoring framework below checks whether each generated protocol makes feasible choices for the method, labeling strategy, refractive-index condition, and timing.

C. Scoring Framework

ClearEval reports two main scores (Table II). The Knowledge Index (I_K) is the score of the static TOC knowledge audit. It macro-averages accuracy across the three MCQ groups: Tissue, Reagent, and Method. I_K is not used to score a generated protocol. It is used to separate lack of basic TOC knowledge from failures in applying that knowledge to protocol design.

The Application Index (I_A) is the protocol-level feasibility score. It evaluates each generated protocol with three dimensions: Completeness, Correctness, and Effectiveness:

$$I_A = \min(\text{Completeness}, \text{Correctness}, \text{Effectiveness}). \quad (1)$$

All three dimensions are reported on a $[0, 100]$ scale. The minimum rule follows wet-lab practice: a protocol is limited by its weakest essential part. A protocol may be detailed and well written, but it is not feasible if it is incomplete, internally inconsistent, or fails an experimental feasibility check. Thus, I_A is the primary score for generated TOC protocols, while I_K is a separate knowledge audit.

a) *Completeness and Correctness.*: Completeness checks whether a protocol contains the information needed to run a TOC experiment. Step Integrity checks whether core stages are present when needed, including fixation or pre-treatment,

TABLE II
METRIC SUMMARY OF CLEARVAL. I_K IS A STATIC TOC KNOWLEDGE
AUDIT. I_A SCORES GENERATED TOC PROTOCOLS. H_{KA} IS A
MODEL-LEVEL SUMMARY THAT COMBINES KNOWLEDGE AND
APPLICATION PERFORMANCE.

Level	Dimension	Sub-Metric	Method
Knowledge (I_K)	Knowledge audit	Accuracy	Deterministic
		Completeness	Step Integrity Parameter Detail
Application (I_A)	Correctness	Step Order (co_order)	LLM rubric
		Method Consistency (co_method)	LLM rubric
		Parameter Consistency (co_param)	LLM rubric
		Chemical Safety (co_chem)	LLM rubric
	Effectiveness	Method Suitability (S_{method})	Signed fit
		Label Compatibility (S_{label})	Rule-based
		RI Matching (S_{trans})	Formula
		Timing (S_{time})	Formula
Diagnostic summary	Knowledge-Application Harmonic Score (H_{KA})	Harmonic mean of I_K and I_A	Aggregation

labeling, RI matching, and imaging preparation. Parameter Detail checks whether key parameters are specified, such as method choice, sample scale, incubation time, temperature, concentration, and labeling conditions.

Correctness checks whether the protocol is internally consistent with TOC practice. It includes four equally weighted sub-checks: Step Order (co_order), whether the wet-lab steps follow a reasonable order; Method Consistency (co_method), whether the selected method fits the goal and sample; Parameter Consistency (co_param), whether scale, timing, concentration, and temperature are consistent; and Chemical Safety (co_chem), whether the protocol introduces an unsafe reagent or operation. Completeness and Correctness are scored by an LLM judge (GPT-5.4) using fixed rubrics, because they require reading free-form procedural text.

b) *Effectiveness.*: Effectiveness checks whether the key experimental choices in a generated protocol are feasible for TOC. It is computed with non-LLM rules and formulas using values from the TOC feasibility database. For each generated protocol, the scorer extracts the clearing method, labeling target, fluorescent marker, and total clearing processing time. These fields are scored by four normalized sub-scores:

$$E = \frac{1}{4} (S_{method} + S_{label} + S_{trans} + S_{time}), \quad (2)$$

where $E \in [0, 1]$. The reported Effectiveness score is $100E$.

Method suitability (S_{method}). Method suitability checks whether the selected clearing method fits the scenario. Each clearing method has a signed six-axis capability vector. The axes describe fluorescent-protein preservation, staining requirement, clearing capacity, sample geometry, operational economy, and safety. Positive values mean the method supports that need; negative values mean the method conflicts with it.

Each scenario has a fixed demand vector prepared from scenario metadata and reviewed before model evaluation. The vector gives each axis a target value $t_a \in [0, 1]$ and a weight $w_a \in \{0, 0.5, 1\}$. A weight of 0 means the scenario does not depend on that axis. For labeling, only the relevant route is emphasized. If fluorescent-protein preservation is needed,

the preservation axis is doubled and the dye axis is ignored; otherwise the reverse is applied. The signed method fit is

$$F_{method} = \frac{\sum_a d_a c_a}{\sum_a d_a} \in [-1, +1], \quad (3)$$

We normalize it to $S_{method} = (F_{method} + 1)/2 \in [0, 1]$ before computing Effectiveness.

Label compatibility (S_{label}). Label compatibility checks whether the proposed labeling strategy matches the biological target and the selected clearing method. The label score is

$$S_{label} = \min(R_{target}, R_{marker}, R_{method-marker}) \in [0, 1]. \quad (4)$$

R_{target} checks whether the marker matches the required labeling target. R_{marker} checks marker–fluorophore compatibility. $R_{method-marker}$ checks whether the fluorophore is compatible with the selected clearing method using the quenching matrix.

Each term is scored on $[0, 1]$. The minimum rule is used because failure in any one part makes the labeling strategy unusable.

RI matching (S_{trans}). RI matching checks whether the selected method fits the tissue and sample scale. The score penalizes the gap between the tissue reference RI (RI_{tissue}) and the method RI (RI_{ref}):

$$S_{trans} = \exp\left(-\frac{(RI_{tissue} - RI_{ref})^2}{2\sigma_{RI}^2}\right) \in [0, 1]. \quad (5)$$

S_{trans} is set to 0 when the selected method has no supported entry for the requested sample scale.

Timing (S_{time}). Timing checks whether the proposed total processing time falls within an evidence-supported window $[t_{min}, t_{max}]$. These windows come from TOC literature, 372 wet-lab experimental records, and expert review. Let Δt be the distance from this window, with $\Delta t = 0$ when the proposed time falls inside the window. The timing score is

$$S_{time} = \exp\left(-\frac{\Delta t^2}{2\tau^2}\right) \in [0, 1], \quad (6)$$

where τ is set to $0.2 \times$ the median clearing time.

All timing windows, RI references, σ_{RI} values, and τ values come from literature, wet-lab records, and expert review. Effectiveness is computed from the four sub-scores and then scaled to $[0, 100]$ before entering the I_A minimum rule.

c) *Knowledge–Application Harmonic Score.*: We report a Knowledge–Application Harmonic Score (H_{KA}) as a model-level summary:

$$H_{KA} = \frac{2I_K I_A}{I_K + I_A}. \quad (7)$$

H_{KA} is a diagnostic summary, not a protocol-feasibility score. It decreases when a model has high TOC knowledge but low protocol-level feasibility.

IV. EXPERIMENTS

A. Models and Settings

We evaluate 13 LLMs across the GPT, Gemini, Claude, DeepSeek, GLM, and Qwen families. All main runs use

TABLE III
AGREEMENT OF AUTOMATED VS. BLINDED EXPERT-CONSENSUS SCORES ON 260 PROTOCOLS (20 SCENARIOS $\times$ 13 MODELS). ICC=ICC(2,1); QWK=QUADRATIC-WEIGHTED κ . FULL 14-ROW BREAKDOWN IN APPENDIX B.

Dimension	Spearman	ICC	QWK
Completeness	0.79	0.75	0.70
Correctness	0.77	0.74	0.72
Effectiveness	0.79	0.80	0.79
CCE Overall	0.82	0.80	0.80
<i>Effectiveness sub-metrics</i>			
S_{method} (method suit.)	0.77	0.82	0.77
S_{label} (rule)	0.76	0.75	0.73
S_{trans} (physics)	0.15	0.64	0.62
S_{time} (physics)	0.75	0.78	0.75

the same task prompts and temperature 0.0 for both the Knowledge audit and the Application track. The Application setting uses a one-shot protocol example as the default prompt. Reasoning variants run in their native thinking mode when available. We release provider snapshot strings, context windows, maximum output tokens, API call dates, prompts, raw responses, and scoring outputs in the repository configuration.

B. Calibration of the Automated Metrics

Before reporting model results, we validate the automated CCE scores against blinded expert consensus on 260 protocols (20 scenarios $\times$ 13 models). We report Spearman ρ , ICC(2,1), and quadratic-weighted κ (QWK) (Table III; full 14-row breakdown in Appendix B).

The aggregate CCE scores agree well with expert consensus. ICC is 0.75 for Completeness, 0.74 for Correctness, and 0.80 for Effectiveness. The overall protocol score also reaches ICC 0.80. Among Effectiveness sub-scores, S_{method} agrees well with expert method-suitability ratings (ICC 0.82). After correction, S_{label} also shows good agreement with expert scores (ICC 0.75), supporting its use in the failure analysis. S_{time} remains well aligned with experts (ICC 0.78). S_{trans} shows moderate absolute agreement (ICC 0.64), although its rank correlation is low; we therefore use RI-related results as a secondary diagnostic.

For Correctness, the aggregate ICC is 0.74. Chemical safety (*co_chem*) is evaluated with a revised rubric that flags introduced hazardous reagents or operations, rather than the absence of unrequested safety caveats. With this rubric, *co_chem* reaches ICC 0.80 and flags hazards in about 28% of protocols.

C. Main Results: Knowledge and Protocol Feasibility

Table IV shows a clear gap between static TOC knowledge and protocol-level feasibility. Knowledge Index scores are high and close across models (79.9–91.1), while Application Index scores are lower and more spread out (35.3–62.3). Fig. 2 shows the same pattern: Knowledge profiles are similar across models, while Application scores show larger differences.

TABLE IV

CLEARVAL RESULTS FOR THE EVALUATED MODELS. I_K IS THE KNOWLEDGE INDEX FROM THE STATIC TOC KNOWLEDGE AUDIT. I_A IS THE APPLICATION INDEX FOR GENERATED PROTOCOLS, DEFINED AS THE MINIMUM OF COMPLETENESS, CORRECTNESS, AND EFFECTIVENESS. H_{KA} IS THE KNOWLEDGE-APPLICATION HARMONIC SCORE, A MODEL-LEVEL DIAGNOSTIC SUMMARY. GLM-4.7-THINK’S THINKING-MODE OUTPUTS OFTEN EXHAUST THE OUTPUT-TOKEN BUDGET AND TRUNCATE ITS PROTOCOLS, LOWERING ITS COMPLETENESS. [†]GEMINI-3-FLASH IS SCORED ON 251/253 SCENARIOS BECAUSE TWO RESPONSES WERE UNPARSEABLE.

Models	Knowledge (MCQ)			Application (OEQ)				H_{KA}
	Tissue	Reagent	Method	I_K	Com.	Cor.	Eff.	I_A
<i>Proprietary Models</i>								
GPT-5.2-Fast	91.1	88.7	93.5	91.1	97.0	84.4	61.2	73.2
GPT-5.2-Think	<u>90.5</u>	<u>87.3</u>	<u>93.6</u>	<u>90.5</u>	96.9	85.4	60.5	72.5
Gemini-3-Flash [†]	78.7	81.8	91.1	83.8	78.2	79.7	59.1	69.3
Gemini-3-Pro	88.8	77.2	86.7	84.2	83.8	<u>85.0</u>	53.1	65.1
Claude-4.6-Sonnet	87.6	82.9	94.9	88.5	98.6	77.4	62.3	<u>73.1</u>
<i>Open-Weight Models</i>								
GLM-4.7	79.3	80.2	93.5	84.3	77.5	73.2	60.6	70.6
GLM-4.7-Think	78.7	70.5	90.7	80.0	51.1	67.2	54.2	62.3
DeepSeek-V3.2	79.9	77.9	91.7	83.1	78.6	68.3	49.3	61.9
DeepSeek-V3.2-Think	85.3	83.8	93.5	87.5	78.2	67.6	49.0	62.9
Qwen3-Max	79.9	<u>87.3</u>	93.1	86.8	87.0	72.5	60.9	71.6
Qwen3-235B-A22B	77.5	<u>81.2</u>	89.1	82.6	76.8	52.4	56.4	64.1
Qwen3-32B	73.4	78.6	89.9	80.6	61.7	50.8	51.3	62.3
Qwen3-14B	73.4	77.8	88.4	79.9	58.4	35.3	53.5	48.9

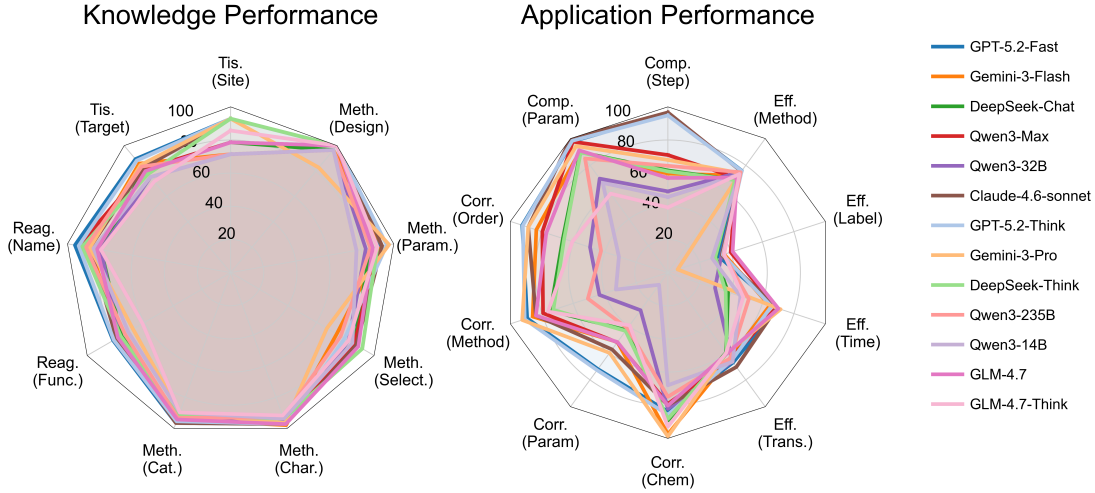


Fig. 2. ClearEval results for 13 LLMs. Left: Knowledge Index components (Tissue, Reagent, Method). Right: Application dimensions and sub-scores. Knowledge profiles are close across models, while Application scores show larger differences.

Models generally score higher on Completeness and Correctness than on Effectiveness. Average Completeness is 79, average Correctness is 69, and average Effectiveness is 56. Effectiveness is the bottleneck for 9 of the 13 models. The remaining models are limited by Correctness or by truncated Completeness. Qwen3-235B-A22B, Qwen3-32B, and Qwen3-14B are limited by Correctness, with Qwen3-14B reaching 35.3 on this dimension. GLM-4.7-Think is limited by Completeness because its reasoning-mode outputs are often truncated.

These results show why the Knowledge audit and the Application score are both needed. A model can answer TOC knowledge questions well but still make infeasible protocol choices. The main feasibility error is label suitability: models

often propose fluorescent markers that do not match the required labeling targets.

D. Failure Analysis

Fig. 3 localizes the main source of low Effectiveness scores. Across all 3,287 scored protocols, label suitability (S_{label}) has the lowest mean score (0.32), while method suitability is higher ($S_{method} = 0.73$). RI matching ($S_{trans} = 0.64$) and timing ($S_{time} = 0.56$) also reduce Effectiveness, but they are less severe than labeling in the aggregate view. Since the corrected S_{label} score agrees well with expert consensus (ICC 0.75), we use it as the main signal for labeling failure.

The per-model S_{label} distributions show that this failure is not limited to one model. Most models have a large mass at

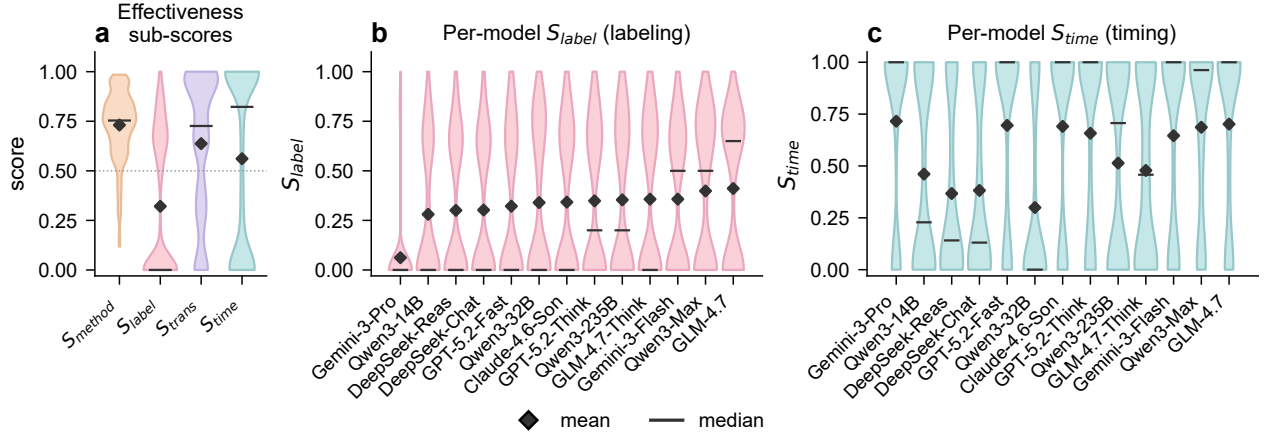


Fig. 3. Failure localization. (A) Effectiveness sub-score distributions over all 3,287 protocols, normalized to $[0, 1]$: S_{label} has the lowest mean (0.32) while S_{method} is high (0.73). (B) Per-model S_{label} distributions (sorted worst-first) (C) Per-model S_{time} distributions.

$S_{label} = 0$, although the average score varies across models. In total, 53% of protocols receive $S_{label} = 0$. Among these zero-score cases, 86% involve marker–target mismatch. These categories are not mutually exclusive, so a protocol may also include marker–fluorophore or method–marker compatibility errors. The main labeling failure is therefore choosing a marker that does not match the requested biological target.

Timing errors form a secondary pattern. The per-model S_{time} distributions are broad, showing that some protocols use reasonable timing while others fall outside evidence-supported windows. These errors often appear in small-sample settings, where models over-estimate processing time. Together, the failure analysis shows that low protocol feasibility is mainly driven by labeling design, with RI matching and timing as additional constraints.

E. Inference-Time Grounding

We run a controlled grounding test on three models: GPT-5.2-Fast, Qwen3-Max, and Qwen3-14B. Each model is evaluated under the one-shot default, one-shot with KB-RAG, and one-shot with KB-RAG plus self-check. KB-RAG provides per-scenario context from scenario metadata, including scenario slots, broad target–marker guidance, and qualitative method and compatibility records. It gives candidate ranges rather than exact protocol answers, and withholds gold protocols, expert scores, exact scoring thresholds, and numeric RI/time answer keys. Self-check applies one fixed feasibility checklist and revises once. All outputs are scored by the same calibrated CCE pipeline.

KB-RAG improves label suitability for all three models, and self-check further improves or preserves this gain. With KB-RAG plus self-check, S_{label} increases from 0.32 to 0.60 for GPT-5.2-Fast, from 0.40 to 0.56 for Qwen3-Max, and from 0.28 to 0.55 for Qwen3-14B. This supports the diagnosis that marker–target matching is an addressable failure mode.

The effect on I_A is smaller and model-dependent. KB-RAG plus self-check improves GPT-5.2-Fast and Qwen3-14B, but not Qwen3-Max, where the label gain is offset by lower RI and timing scores. Since these changes are not tested for statistical

TABLE V
INFERENCE-TIME GROUNDING ON THREE MODELS OVER 253 APPLICATION SCENARIOS. COM/COR/EFF/ I_A ARE PERCENTAGES. EFFECTIVENESS SUB-SCORES ARE NORMALIZED MEANS IN $[0, 1]$. **BOLD**=BEST VALUE PER COLUMN WITHIN EACH MODEL. KB-RAG WITHHOLDS GOLD PROTOCOLS AND NUMERIC RI/TIME ANSWER KEYS. SELF-CHECK IS ONE CHECKLIST-BASED REVISION.

Model	Setting	Com	Cor	Eff	I_A	S_m	S_{lab}	S_{tr}	S_{ti}
GPT-5.2-Fast	1-shot	97.0	84.4	61.2	61.2	0.76	0.32	0.67	0.70
	+KB-RAG	94.2	79.0	62.3	62.3	0.79	0.45	0.67	0.57
	+KB-RAG+SC	92.0	80.2	63.2	63.2	0.79	0.60	0.60	0.54
Qwen3-Max	1-shot	87.0	72.5	60.9	60.9	0.71	0.40	0.64	0.69
	+KB-RAG	85.1	73.4	57.8	57.8	0.77	0.50	0.56	0.49
	+KB-RAG+SC	85.2	73.6	58.6	58.6	0.76	0.56	0.57	0.45
Qwen3-14B	1-shot	58.4	35.3	53.5	35.3	0.73	0.28	0.66	0.46
	+KB-RAG	63.1	41.8	59.8	41.8	0.73	0.52	0.66	0.48
	+KB-RAG+SC	63.3	42.4	60.6	42.4	0.73	0.55	0.66	0.49

significance, we interpret this experiment as a diagnostic grounding probe rather than a full prompting comparison. Overall, grounding helps labeling, but full protocol feasibility still depends on method, RI, and timing choices.

Retrieval variants. We also tested three ways of building the retrieved context: (i) candidate clearing methods ranked by how well they fit the scenario’s needs; (ii) methods ranked by whether their refractive index matches the tissue; and (iii) the same refractive-index ranking plus a rough fast/slow speed hint for each method. These variants changed different Effectiveness sub-scores. Label compatibility improved across variants, while RI and timing scores changed with the retrieved method context. None of the variants improved all four Effectiveness sub-scores at the same time.

We therefore report the released, non-per-model-tuned KB-RAG setting as a controlled grounding test. These results suggest that grounding context affects the balance among method, labeling, RI, and timing. Joint optimization of these choices is outside the scope of this benchmark study.

V. DISCUSSION AND CONCLUSION

ClearEval shows that static TOC knowledge does not reliably lead to feasible protocol generation. In our experiments,

the main measured failure is marker–target matching in labeling design. RI matching and timing also affect protocol feasibility. These findings support the need for protocol-level checks grounded in experimental rules, rather than evaluation based only on knowledge accuracy or text quality.

Scope. ClearEval is a TOC-specific benchmark. Its claims are about TOC protocol feasibility, not general LLM scientific competence. Applying the same approach to another wet-lab domain would require new domain evidence, expert-reviewed rules, and task scenarios. CCE should therefore be viewed as a re-curatable evaluation template, not a finished multi-domain benchmark. ClearEval also complements broad protocol benchmarks such as BioProBench [12]: those benchmarks cover more biological areas and task types, while ClearEval provides a deeper feasibility check within one subdomain.

Limitations. ClearEval is an offline pre-experimental benchmark: it does not model all wet-lab variation (reagent age, batch effects, tissue heterogeneity, hardware), and its database is literature-, record-, and expert-based rather than exhaustively re-measured. Its calibrated CCE scores do not replace real wet-lab validation, and the grounding experiment is a controlled KB-RAG/self-check test, not a full search over prompting, retrieval, or agent settings.

Overall, ClearEval turns expert judgment on TOC protocol feasibility into computable checks over method choice, labeling, tissue, and timing. It provides a depth-first feasibility benchmark for TOC and a re-curatable template for building similar checks in other wet-lab domains.

APPENDIX A

KNOWLEDGE-BASE CURATION AND PROVENANCE

Expert review. The TOC feasibility database and CCE rubrics were curated with input from tissue-clearing wet-lab experts and a knowledge-graph specialist. The review focused on method properties, tissue and sample-scale settings, labeling targets, marker compatibility, refractive-index matching, timing windows, and safety-related rules.

Knowledge bases and provenance. Seven JSON knowledge bases (Table I) support the Effectiveness checks, storing method capability vectors, marker/fluorophore compatibility, tissue and target information, RI references, timing windows, wet-lab records, and literature chunks.

APPENDIX B

EXPERT PANEL AND SCORING PROCEDURE.

The reference scores come from a two-stage expert-panel procedure, not a single annotator. First, a wet-lab TOC expert scored every protocol blind to model identity, using the same CCE rubric applied by the automated pipeline. A panel of five TOC wet-lab experts and one knowledge-graph specialist then reviewed and adjusted these scores through discussion, combining their complementary backgrounds to adjudicate hard cases such as incubation-time scaling and marker–fluorophore compatibility. This consensus procedure improves consistency and avoids the failure modes of both single-rater scoring and independent multi-rater majority voting; because it yields one

consensus reference per protocol, we report automated-vs-consensus agreement rather than human–human reliability.

TABLE VI
FULL PER-DIMENSION AGREEMENT OF AUTOMATED VS.
EXPERT-CONSENSUS SCORES ($n=260$). ICC=ICC(2,1);
QWK=QUADRATIC-WEIGHTED κ .

Dimension	ρ	ICC	QWK
<i>c_step</i>	0.72	0.70	0.54
<i>c_param</i>	0.71	0.68	0.60
Completeness	0.79	0.75	0.70
<i>co_order</i>	0.65	0.61	0.58
<i>co_method</i>	0.81	0.79	0.77
<i>co_param</i>	0.56	0.53	0.48
<i>co_chem</i>	0.69	0.80	0.66
Correctness	0.77	0.74	0.72
<i>S_method</i>	0.77	0.82	0.77
<i>S_label</i>	0.76	0.75	0.73
<i>S_trans</i>	0.15	0.64	0.62
<i>S_time</i>	0.75	0.78	0.75
Effectiveness	0.79	0.80	0.79
Overall	0.82	0.80	0.80

APPENDIX C

WORKED SCORING EXAMPLE

To make CCE concrete, Table VII traces one real Application item end to end (GPT-5.2-Fast, one-shot): the dual-target embryo-clearing scenario, the protocol the model generates, and its CCE scoring. Completeness and Correctness are perfect, and method fit, RI matching, and timing score highly; yet $S_{label}=0$, because the reply labels the early-neuron target correctly (TUJ1) but substitutes Cntn2 for the recommended Neurofilament marker. Since S_{label} is a minimum over the required targets, this single off-target marker zeroes it, dragging I_A to 0.72 despite perfect Completeness and Correctness.

REFERENCES

- [1] H. R. Ueda, A. Ertürk, K. Chung *et al.*, “Tissue clearing and its applications in neuroscience,” *Nature Reviews Neuroscience*, vol. 21, no. 2, pp. 61–79, 2020.
- [2] K. Chung and K. Deisseroth, “Structural and molecular interrogation of intact biological systems,” *Nature*, vol. 497, no. 7449, pp. 332–337, 2013.
- [3] D. Jin, E. Pan, N. Oufattole *et al.*, “What disease does this patient have? a large-scale open domain question answering dataset from medical exams,” *Applied Sciences*, vol. 11, no. 14, p. 6421, 2021.
- [4] D. Hendrycks, C. Burns, S. Basart *et al.*, “Measuring massive multi-task language understanding,” in *International Conference on Learning Representations*, 2021.
- [5] Q. Chen *et al.*, “Benchmarking large language models for biomedical natural language processing applications and recommendations,” *Nature Communications*, vol. 16, p. 3280, 2025.
- [6] O. Bianchi *et al.*, “CARDBiomedBench: a benchmark for evaluating the performance of large language models in biomedical research,” *The Lancet Digital Health*, 2025.
- [7] S. S. Li *et al.*, “MediQ: Question-asking LLMs and a benchmark for reliable interactive clinical reasoning,” in *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024.
- [8] K. Papineni, S. Roukos, T. Ward *et al.*, “BLEU: a method for automatic evaluation of machine translation,” in *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, 2002, pp. 311–318.

TABLE VII

WORKED CCE EXAMPLE (GPT-5.2-FAST, ONE-SHOT): THE SCENARIO, THE GENERATED PROTOCOL, AND THE CCE BREAKDOWN FOR ONE REAL APPLICATION ITEM. COMPLETENESS/CORRECTNESS SUB-SCORES ARE RUBRIC POINTS (RAW/MAX); EFFECTIVENESS SUB-SCORES ARE NORMALIZED TO $[0, 1]$ ($S_{method}=(F+1)/2$). VALUES ARE FROM THE RELEASED SCORER OUTPUT.

OEQ prompt . “Clear an E12.5 mouse embryo ($\sim 8\text{--}10\text{ mm}$, fixed) to simultaneously visualize the more mature, bundled cranial/spinal nerves and the extending/branching early neurites entering peripheral targets, so as to analyze how the embryonic nervous system builds central-to-peripheral connectivity in 3D. Choose a compatible <i>dual-target</i> labeling strategy and a compatible clearing method suited to whole-embryo neural labeling, low-background deep imaging, continuous tract tracing, and early-neurite visualization. Output only the protocol (line 1: the chosen labeling strategy and clearing method), then the key reagents, conditions, and times for clearing, RI matching, and storage/pre-imaging.” <i>Answer space</i> : one clearing method from a fixed set of 18; fluorophores/markers from fixed lists. <i>Database-recommended markers</i> : Neurofilament (mature bundles), β -III Tubulin/TUJ1 (early neurites).			
Model response <i>Method</i> : iDISCO+. <i>Labeling</i> : TUJ1+Alexa 647, Cntn2+Alexa 568. <i>Justification</i> : “iDISCO+ suits deep, low-background light-sheet imaging of an intact embryo; MeOH dehydration + DCM delipidation give high transparency and long-range tract tracing; TUJ1 marks early neurites, Cntn2 mature bundled axons, and Alexa 647/568 stay stable after organic-phase RI matching.” <i>Protocol</i> : 1 dehydration — PBS wash (RT, $3\times 20\text{ min}$), then 50/80/100% and fresh 100% MeOH (RT, 1 h each); 2 bleaching — 5% $\text{H}_2\text{O}_2/\text{MeOH}$ (4°C , 12–16 h), MeOH wash; 3 rehydration/permeabilization — 80/50% MeOH, PBS, then 0.2% Triton/20% DMSO/0.3 M glycine and block (10% DMSO/6% donkey serum) (37°C , 12–16 h each); 4 primary — anti-TUJ1+anti-Cntn2 (1:200, 37°C , 4–6 d); 5 wash (RT then 37°C); 6 secondary — Alexa 647/568 (1:300, 37°C , 3–5 d); 7 wash; 8 re-dehydration (50–100% MeOH); 9 delipidation — 66% DCM/MeOH then 100% DCM (RT); 10 RI matching — DBE ($\text{RI}\approx 1.56$), 2–3 h then fresh DBE 12–24 h; 11 storage/mounting in DBE and light-sheet imaging at $\text{RI}\approx 1.56$ (Alexa 568/647).			
Dimension	Sub-metric	Value	Rationale (released scorer output)
Completeness	Step modules c_{step}	2/2	All iDISCO+ modules present: dehydration, bleaching, permeabilization/blocking, dual immunostaining, re-dehydration, DCM delipidation, DBE RI matching, mounting.
	Parameters c_{param}	3/3	Concentration, temperature, and time are given for essentially every step.
Correctness	Order co_{order}	3/3	Canonical iDISCO+ sequence: dehydrate/bleach $\rightarrow$ rehydrate/permeabilize $\rightarrow$ stain $\rightarrow$ re-dehydrate $\rightarrow$ delipidate $\rightarrow$ RI match.
	Method co_{method}	2/2	MeOH/DCM/DBE reagents mutually consistent; no aqueous-RIMS contamination.
	Parameters co_{param}	2/2	4–6 d primary / 3–5 d secondary staining and 12–24 h DBE are feasible at whole-embryo scale.
	Chem/safety co_{chem}	1/1	Standard reagents; a solvent-ventilation caveat is noted but not penalized.
Effectiveness	Method fit S_{method}	0.96	iDISCO+ strongly fits a whole embryo: signed need-weighted fit $F=0.91$, rescaled $(F+1)/2=0.96$.
	Label S_{label}	0.00	$\min(R_{target}=0, R_{marker}=1.00, R_{method-marker}=0.85)=0$. The 0 is R_{target} : TUJ1 hits the neurite target, but the nerve-bundle target (recommended <i>Neurofilament</i>) is given <i>Cntn2</i> , which matches no required target; target-match is a min over committed markers, so one off-target marker zeroes it. ($R_{method-marker}=0.85$: Alexa 568 slightly less solvent-stable in iDISCO+, non-binding.)
	RI match S_{trans}	0.92	Tissue RI 1.48 vs. method reference 1.47 ($\sigma_{RI}=0.025$).
	Timing S_{time}	1.00	The <i>clearing-phase</i> time only (dehydration+delipidation+RI matching; multi-day antibody staining is labeling, excluded): $\approx 36\text{ h}$, inside the iDISCO+ window [12, 48] h.
Com =1.00, Cor =1.00, Eff = $\frac{1}{4}(0.96+0+0.92+1.00)=0.72$, $I_A=\min(\text{Com}, \text{Cor}, \text{Eff})=\mathbf{0.72}$			

- [9] Y. Liu, D. Iter, Y. Xu, S. Wang, R. Xu, and C. Zhu, “G-Eval: NLG evaluation using GPT-4 with better human alignment,” in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 2511–2522.
- [10] J. Gu *et al.*, “A survey on LLM-as-a-Judge,” *arXiv preprint arXiv:2411.15594*, 2024.
- [11] K. Singhal *et al.*, “Large language models encode clinical knowledge,” *Nature*, vol. 620, no. 7972, pp. 172–180, 2023.
- [12] Y. Liu, L. Lv, X. Zhang, J. Wang, L. Yuan, and Y. Tian, “BioProBench: Comprehensive dataset and benchmark in biological protocol understanding and reasoning,” *arXiv preprint arXiv:2505.07889*, 2025.
- [13] J. M. Laurent *et al.*, “LAB-Bench: Measuring capabilities of language models for biology research,” *arXiv preprint arXiv:2407.10362*, 2024.
- [14] L. Mitchener *et al.*, “BixBench: a comprehensive benchmark for LLM-based agents in computational biology,” *arXiv preprint arXiv:2503.00096*, 2025.
- [15] J. Ying *et al.*, “SeedBench: A multi-task benchmark for evaluating large language models in seed science,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025, pp. 31 395–31 449.
- [16] L. Zheng *et al.*, “Judging LLM-as-a-Judge with MT-Bench and chatbot arena,” in *Advances in Neural Information Processing Systems 36 (NeurIPS 2023) Datasets and Benchmarks Track*, 2023.
- [17] Y. Deng, J. Xu, T. Yu, and D. Zhu, “Advances in tissue optical clearing for 3D imaging in large animal,” *Frontiers of Optoelectronics*, vol. 18, no. 3, p. 18, 2025.
- [18] E. A. Susaki, “Unlocking the potential of large-scale 3D imaging with tissue clearing techniques,” *Microscopy*, vol. 74, no. 3, pp. 179–188, 2025.
- [19] H. Mai and D. Lu, “Tissue clearing and its applications in human tissues: A review,” *VIEW*, vol. 5, no. 2, p. 20230046, 2024.